

25 July 2026

Editorial Office

Expert Systems with Applications

Re: Research article submission, “ContractRAG: A risk-certified expert system for deploying retrieval-augmented generation services”

Dear Editor,

On behalf of all authors, we submit this manuscript for consideration as a Research Article in *Expert Systems with Applications*.

Organizations deploying retrieval-augmented generation (RAG) must choose among many retrieval, reranking, and generation configurations. Current systems usually recommend a configuration from average development-set performance. They do not tell an operator how often the deployed service may return a wrong, unsupported, stale, or late answer. Our manuscript presents ContractRAG, a risk-aware expert system that converts such operational requirements into an auditable deployment decision.

A domain expert first defines an evidence contract containing measurable losses, admissible violation rates, and a confidence budget. ContractRAG then compiles heterogeneous knowledge-access plans, learns progressive policies that acquire stronger evidence only when needed, and uses a distribution-free test to recommend the least-cost policy in a certified prefix. When no candidate can meet the contract, the system reports infeasibility instead of silently weakening the requirement. After deployment, an anytime-valid monitor detects drift and triggers a fresh decision cycle. The result is a complete design–test–deploy–monitor workflow rather than a stand-alone routing component.

We evaluate this workflow on HybridQA, CRAG, ASQA, and QAMPARI using four large language model families and two serving backends. Across 1,000 calibration draws, recommended policies exceed their contracts in at most 0.8% of draws under a 10% error budget, compared with up to 64% for empirical and Bayesian tuning. They reduce model cost by up to $20\times$, or $10\text{--}18\times$ after audit cost. The evaluation also covers strict and joint contracts, human deferral, infeasible requirements, audit sampling, and post-deployment drift.

The work fits the journal’s focus on the design, development, testing, and management of expert and intelligent systems. It supplies an operator-facing decision process for information retrieval and risk assessment, an implemented architecture, formal guarantees, and practical evidence across several public tasks.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and its submission. The authors declare no competing financial interests or personal relationships that could have influenced the work.

Thank you for considering our submission.

Sincerely,

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